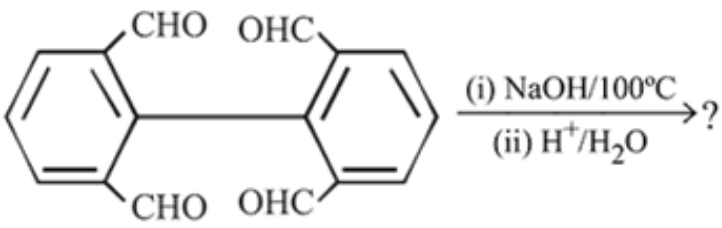
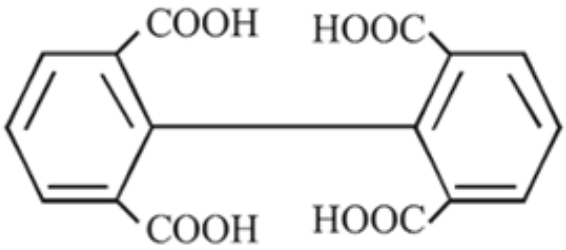
Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

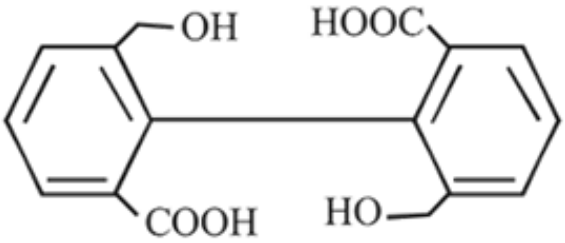


The major product in the above reaction is

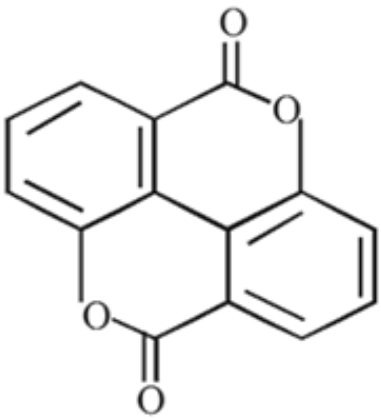
(A)



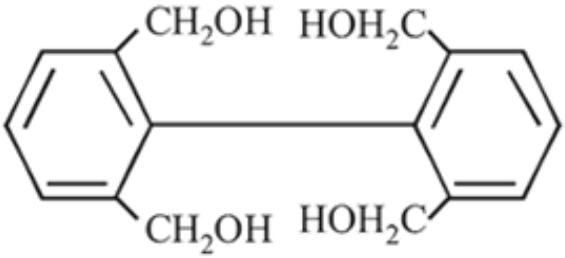
(B)



(C)



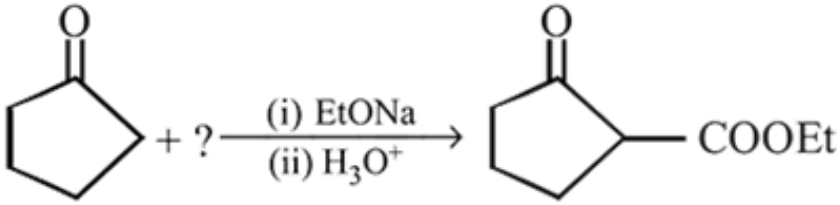
(D)



@IITJEE_Advanced

Q.2

Identify the missing reagent in the reaction given below



(A) Diethyl carbonate $(EtO)_2C = O$

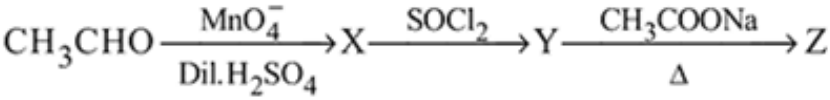
(B) Ethyl acetate CH_3COOEt

(C) Ethyl formate $HCOOEt$

(D) Diethyl oxalate Et. OOC. COOEt

Q.3

Identify Z in the following reaction sequence.



(A) CH_3COCH_2COONa

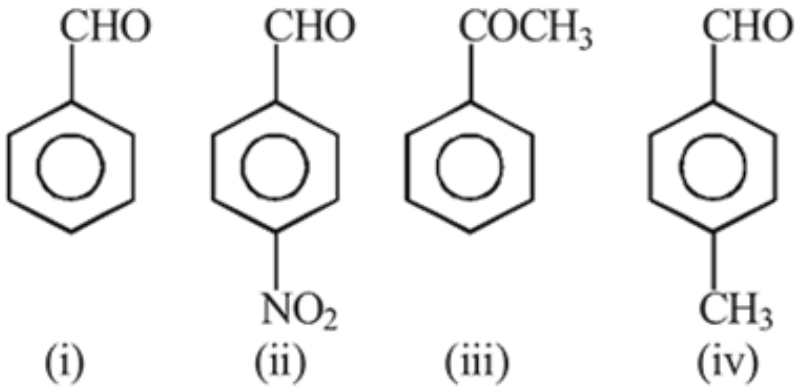
(B) $(CH_3CO)_2O$

(C) $CH_3CO - O - COCH_2Cl$

(D) $CH_3CO - OCOCHCl_2$

Q.4

Arrange the following carbonyl compounds in decreasing order of their reactivity in nucleophilic addition reaction.



(A) (ii) $>$ (iii) $>$ (i) $>$ (iv)

(B) (ii) $>$ (i) $>$ (iv) $>$ (iii)

(C) (iii) $>$ (ii) $>$ (i) $>$ (iv)

(D) (iii) $>$ (i) $>$ (iv) $>$ (ii)

@IITJEE_Advanced

Q.5

Oxidation of acetaldehyde with selenium dioxide produces

(A) ethanoic acid

(B) methanoic acid

(C) glyoxal

(D) oxalic acid

Q.6

Which of the following will respond to Cannizzaro's reaction?

(A) 2,2-Dimethylpropanal

(B) Acetaldehyde

(C) Propionaldehyde

(D) Cinnamaldehyde

Q.7

If 3 -hexanone is reacted with $NaBH_4$ followed by hydrolysis with D_2O, the product will be

(A) $CH_3CH_2CH(OH)CH_2CH_2CH_3$

(B) $CH_3CH_2CD(OH)CH_2CH_2CH_3$

(C) $CH_3CH_2CH(OD)CH_2CH_2CH_3$

(D) $CH_3CH_2CD(OD)CH_2CH_2CH_3$

Q.8

Propanal on reaction with dilute $NaOH$ forms

(A) $CH_3CH_2CH_2CH_2CH_2CHO$

(B) $CH_3CH_2CH(OH)CH_2CH_2CHO$

(C) $CH_3CH_2CH_2CH(OH)CH_2CHO$

(D) $CH_3CH_2CH(OH)CH\left(CH_3\right)CHO$

Q.9

Among the given compounds, the most susceptible to nucleophilic attack at the carbonyl group is

(A) $MeCOCl$

(B) $MeCHO$

@IITJEE_Advanced

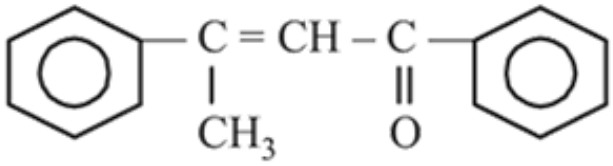
(C) $MeCOOMe$

(D) $MeCOOCOMe$

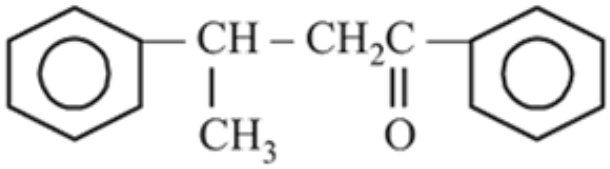
Q.10

Acetophenone when reacted with a base, C_2H_5 ONa, yields a stable compound which has the structure.

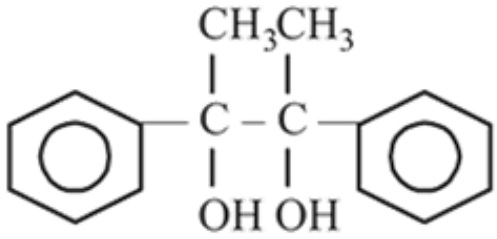
(A)



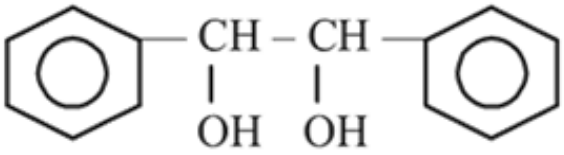
(B)



(C)



(D)



Q.11

Which of the following will react at the slowest rate with a nucleophile

(A) methanal

(B) butanone

(C) 2-pentanone

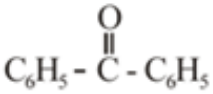
(D) 3-pentanone

Q.12

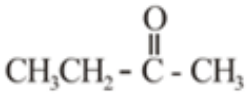
Which of the following gives aldol condensation reaction

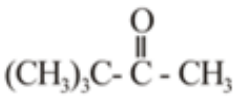
(A) C_6H_5OH

(B)



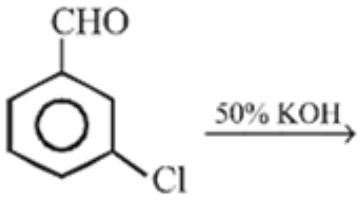
@IITJEE_Advanced

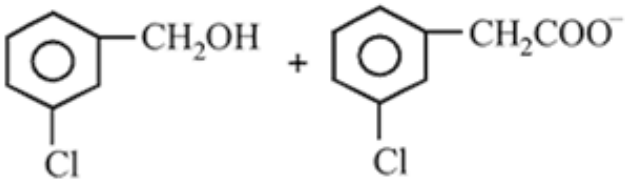
(C) 

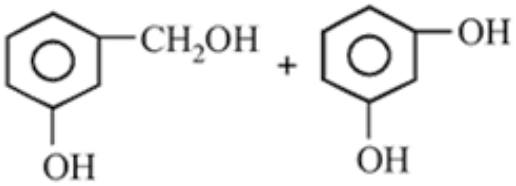
(D) 

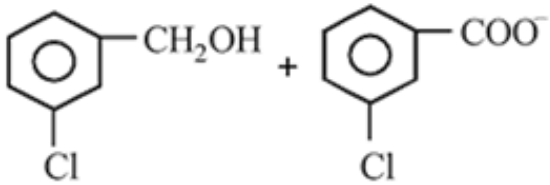
Q.13

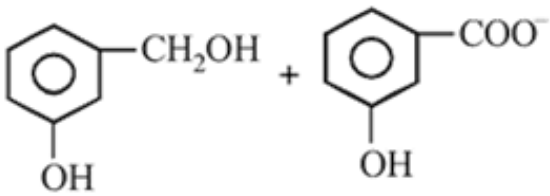
Predict the product in the given reaction.



(A) 

(B) 

(C) 

(D) 

Q.14

Phenylmethyl ketone can be converted into ethylbenzene in one step by which of the following reagents?

(A) $LiAlH_4$

(B) $Zn - Hg/HCl$

(C) $NaBH_4$

(D) CH_3MgI

Q.15

Match the compounds given in List-I with List-II and select the suitable option given below:

@IITJEE_Advanced

List I List-II \ (A) Benzaldehyde (i) Phenolphthalein

(B) Phthalic anhydride (ii) Benzoin condensation

(C) Phenyl benzoate (iii) Oil of wintergreen

(D) Methyl salicylate (iv) Fries rearrangement

(A) (A) - (iv), B - (i), C - (iii), D - (ii)

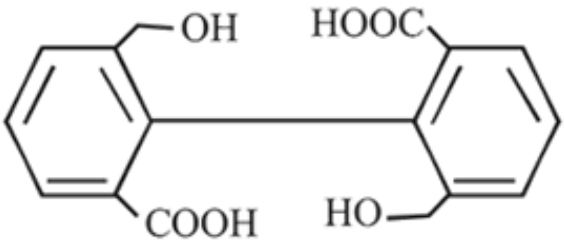
(B) (A) - (iv), B - (ii), C - (iii), D - (i)

(C) (A) - (ii), B - (iii), C - (iv), D - (i)

(D) (A) - (ii), B - (i), C - (iv), D - (iii)

Answers & Solutions

Q.1 Answer:



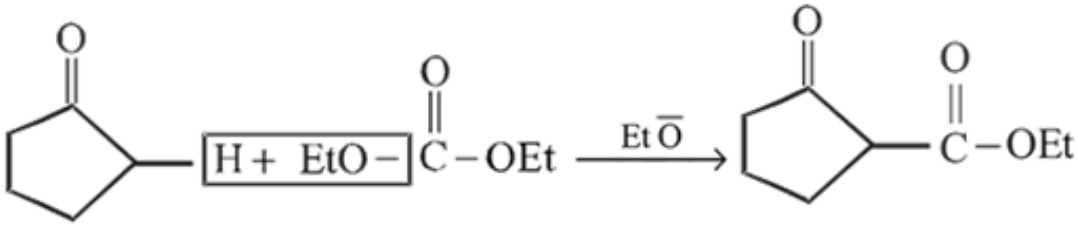
Solution:

This is intramolecular Cannizzaro's reaction.

Q.2 Answer:

Diethyl carbonate $(EtO)_2C=O$

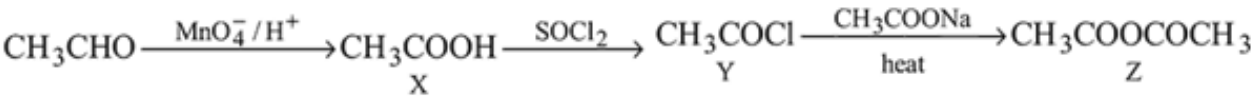
Solution:



Q.3 Answer:

$(CH_3CO)_2O$

Solution:



Q.4 Answer:

$(ii) > (i) > (iv) > (iii)$

Solution:

@IITJEE_Advanced

Aldehyde will be more reactive than ketone.

Q.5 Answer:

glyoxal

Solution:

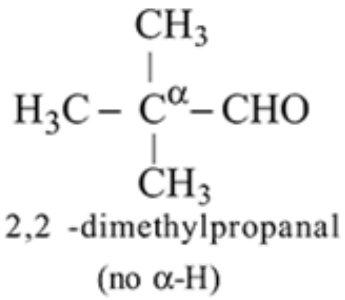
Selenium dioxide oxidises aldehydes and ketones at α -position to carbonyl group. $CH_3CHO \xrightarrow{SeO_2} OHC.CHO$

Q.6 Answer:

2,2-Dimethylpropanal

Solution:

(a)



2,2 -dimethylpropanal

(no α-H)

(b) $H_3C^{\alpha} - CHO$

acetaldehyde $(\alpha - H)$

(c)

$CH_3\overset{\alpha}{C}H_2CHO$

propionaldehyde

(α-H)

(d)

$C_6H_5CH = \overset{\alpha}{C}HCHO$

cinnamaldehyde

(α-H)

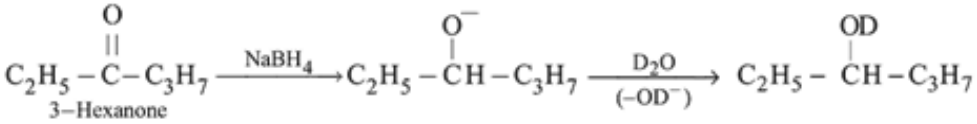
Q.7 Answer:

$CH_3CH_2CH(OD)CH_2CH_2CH_3$

Solution:

@IITJEE_Advanced

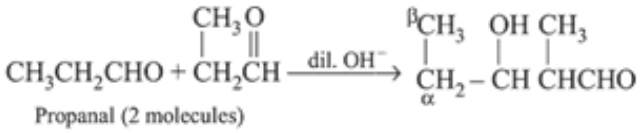
Remember that during reduction with $NaBH_4$, a hydride ion is transferred by $NaBH_4$ to carbonyl carbon and a proton (or deutron) is transferred from the solvent (in the second step) to carbonyl oxygen.



Q.8 Answer:

$CH_3CH_2CH(OH)CH(CH_3)CHO$

Solution:



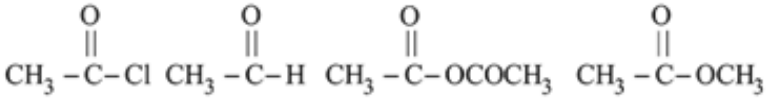
Note that it is the α -carbon (and not β -) that is adding on the carbonyl oxygen of the other propanal molecule.

Q.9 Answer:

$MeCOCl$

Solution:

More the magnitude of positive charge on the carbonyl carbon, higher will be its reactivity toward nucleophilic attack.

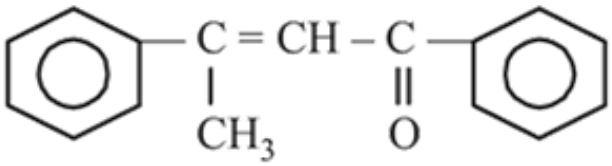


Chlorine being more electron-withdrawing than O develops more positive charge on carbonyl carbon. The order of reactivity of the various carboxyl derivatives toward nucleophile is

$RCOCl > RCHO > (RCO)_2O > RCOOR$

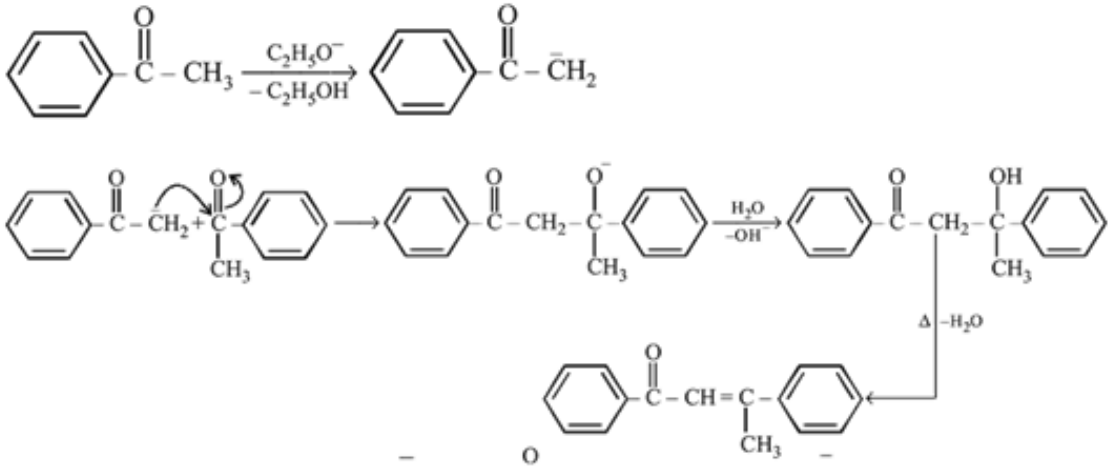
Most reactive **Least Reactive**

Q.10 Answer:



Solution:

@IITJEE_Advanced



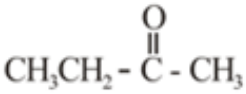
Q.11 Answer:

3-pentanone

Solution:

If the electrophilic character of carbon is considerably reduced, then the reaction will be slow.

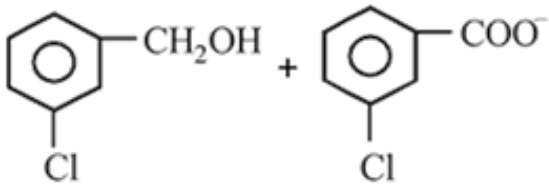
Q.12 Answer:



Solution:

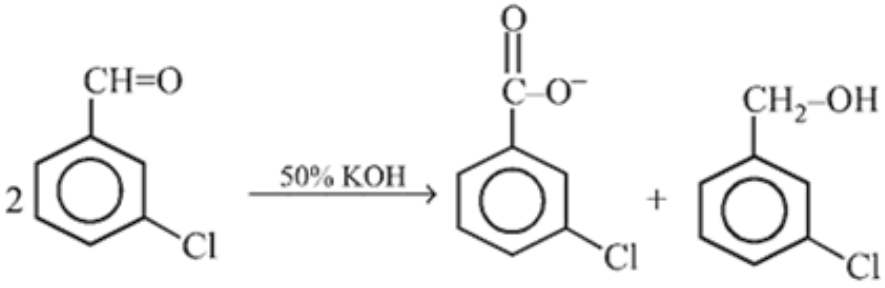
Although both $CH_3CH_2COCH_3$ and $(CH_3)_3\,CCOCH_3$ contain α -hydrogen, yet $(CH_3)_3\,CCOCH_3$ does not undergo Aldol condensation due to steric hindrance

Q.13 Answer:



Solution:

Cannizzaro reaction - when an aldehyde containing no $\alpha - H$ undergo reaction in presence of $50\%KOH$. It disproportionates to form a molecule of carboxylic acid and a molecule of alcohol.

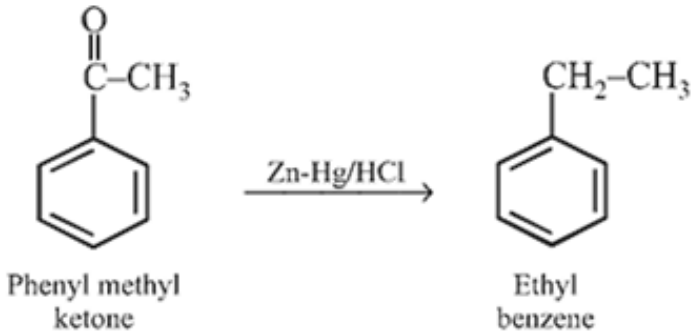


@IITJEE_Advanced

Q.14 Answer:

$Zn - Hg/HCl$

Solution:



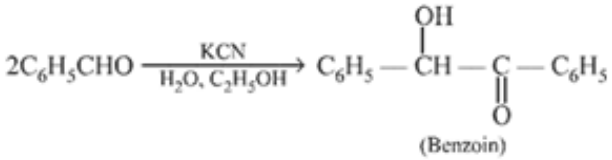
This reaction is known as Clemmensen's reduction.

Q.15 Answer:

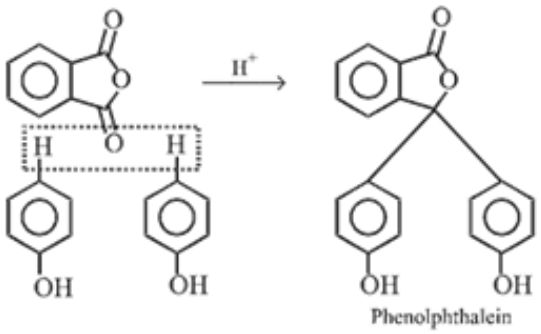
(A) - (ii), B - (i), C - (iv), D - (iii)

Solution:

(a)

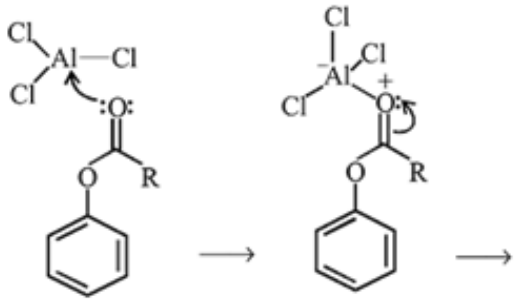


(b)

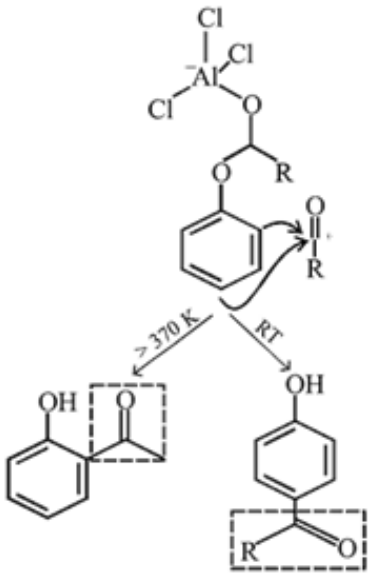


@IITJEE_Advanced

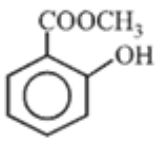
(c)



Fries rearrangement



(d)



(Oil of wintergreen)

@IITJEE_Advanced